

Project Identification

Project Title	Bangla Publishing Studio - A Free, Offline-Ready Web Platform for MS-Word-Like Bengali Book Typesetting with Print-Press Imposition
Scientific Field	Engineering & Innovation
Project Type	Individual Project
Participant	Topu Biswas
Current Class / Year	B.Sc. (Engg.) 2nd Year
Institution	Department of Information and Communication Engineering (ICE), University of Rajshahi
Division / District	Rajshahi / Rajshahi
Age Group	Group C (18 years or above)
Event	ZRF Science Fair 2026 - Ziaur Rahman Foundation

5 Paper sizes incl. Demy & Crown	1-Click Farma (imposition) PDF for press	100% Free, zero installation	Offline PWA autosave, no internet needed
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Key capabilities of Bangla Publishing Studio at a glance.

1. Project Abstract

Publishing a book in Bangladesh still depends on expensive desktop-publishing software, scarce and costly typesetting operators, and repeated proof prints at the press. For teachers, coaching centres, rural writers, and small publishers, these barriers make book production slow, risky, and expensive. **Bangla Publishing Studio** removes these barriers with a free, zero-installation web platform that works as simply as Microsoft Word but produces real, print-ready books in Bengali.

Running entirely in the browser, the platform offers true paginated book layout on standard physical paper sizes (A4, Letter, Legal, Demy, and Crown), native Bengali fonts (Kalpurush, SolaimanLipi, Siyam Rupali), and rich WYSIWYG editing with tables, images, text boxes, shapes, icons, and page headers and footers. Progressive Web App technology keeps the editor fully usable without internet, and an automatic IndexedDB autosave protects every keystroke. Its most distinctive capability is **Farma** generation: the industry-standard imposition scheme used by Bangladeshi printing presses, in which book pages are rearranged into printer's signatures for offset printing. With one click, the same document can be exported as a press-ready Farma PDF, a normal print PDF, a DOCX file, or a self-contained HTML document.

The platform is built with modern web technologies (Next.js, React, and ProseMirror) and performs all processing on the user's own device, keeping it fast, private, and free of server costs. By uniting familiar word-processing simplicity with professional print engineering, Bangla Publishing Studio aims to democratize book production for the millions of Bengali readers, writers, and educators who have been excluded from digital publishing until now.

2. Objectives

1. To develop a free, browser-based WYSIWYG editor that feels as familiar as Microsoft Word but automatically paginates content into real book pages.
2. To implement true print pagination on the standard physical book sizes used in Bangladesh, including A4, Letter, Legal, Demy, and Crown.
3. To generate industry-standard **Farma** (imposition) output with signature-based page ordering and recto-verso pairing, so any local press can print directly without costly desktop-publishing software or manual page rearrangement.
4. To guarantee offline-first reliability through PWA service workers and automatic local autosave, so unstable internet connections never destroy a writer's work.
5. To provide one-click export from a single source file to print-ready PDF, imposed Farma PDF, DOCX, self-contained HTML, and JSON backup.
6. To keep the entire platform completely free and accessible to students, teachers, coaching centres, writers, and small publishers everywhere in Bangladesh.

3. Innovation / Novelty

The core novelty of Bangla Publishing Studio is that it is, to the best of our knowledge, the first browser-based platform that combines three things which have never been united in one free tool: (a) MS-Word-like WYSIWYG editing, (b) true physical-size book pagination, and (c) the Farma imposition scheme of the Bangladeshi printing industry. International software such as Adobe InDesign or MS Publisher ignores both Demy/Crown paper sizes and local imposition practice, forcing Bangladeshi presses to redo layout work manually. This project speaks the exact language of the local press.

- **Zero-setup architecture:** every processing step happens client-side (Next.js + React + ProseMirror). Nothing is installed, nothing is licensed, and nothing is sent to a server - privacy and speed are achieved by design.
- **Offline-first PWA:** after the first visit, the complete editor, Bengali fonts, and autosave work without any internet connection - a decisive advantage for rural users with unstable connectivity.

- **Print-industry awareness:** signature-based Farma sheets with correct recto-verso pairing let a press print, fold, and bind a book exactly as it does today, eliminating an entire class of manual errors.
- **Democratization of publishing:** the entry barrier for producing a properly typeset, press-ready book drops from thousands of taka (software licence plus operator fees) to zero.
- **Local-language engineering:** Bengali fonts and complex-script rendering are treated as first-class features, not an afterthought.

4. Expected Impact

Educational impact. Teachers and coaching centres will be able to self-produce textbooks, guides, and question banks in hours instead of weeks, and university students will gain a working example of modern web engineering - progressive web apps, offline-first storage, and browser-based print technology - that they can study, reuse, and extend for other South Asian languages.

Social and economic impact. Rural and low-income writers will be able to publish novels, poetry, and research without middlemen, because the platform removes software costs, operator fees, and the longest part of the pre-press cycle. Small publishers will cut typesetting costs and turnaround time, keeping more of the value of publishing inside local communities.

Cultural impact. By making Bengali book production as easy as typing a document, the project strengthens the creation and preservation of Bengali-language knowledge in the digital era - more textbooks, more literature, and more local history printed in the mother tongue.

Scalability. The architecture is deliberately open: future versions will add more imposition schemes, professional book templates, collaborative editing, and AI-assisted layout, and the same model can be adapted to other underserved languages with minimal change.

5. Technical Overview

Layer	Technology
Frontend framework	Next.js 16 (App Router), React 19, TypeScript 5, Tailwind CSS
Editing engine	ProseMirror / TipTap with custom nodes for images, text boxes, shapes, and icons
Pagination engine	Client-side true-page layout with physical paper sizes (A4, Letter, Legal, Demy, Crown)
Imposition engine	Farma (signature-based) page reordering with recto-verso pairing for offset printing
Storage	IndexedDB autosave via Dexie.js, plus JSON project backup and restore
Offline support	Progressive Web App with service worker; installable and fully usable offline
Export formats	Print PDF, Farma imposition PDF, DOCX, self-contained HTML, JSON

Layer	Technology
Bengali typography	Kalpurush, SolaimanLipi, and Siyam Rupali font families with complex-script rendering

Table 1: Technology stack of Bangla Publishing Studio.

6. Current Status & Roadmap

A fully working prototype of Bangla Publishing Studio is already deployed on the public internet and is being used to create real documents. The project is version-controlled on GitHub with a clean, audited codebase, continuous quality checks, and a production deployment pipeline. Core features - WYSIWYG editing, true pagination, headers and footers, offline autosave, and PDF, DOCX, HTML, and JSON export - are operational today.

The development roadmap prioritizes: (1) advanced typography controls such as word and letter spacing, paragraph gaps, indents, and drop caps tuned for Bengali conjuncts; (2) refinements of the Farma imposition engine to cover every signature format used by Bangladeshi presses; (3) per-page header and footer overrides for professional front matter; (4) clickable hyperlinks inside exported PDFs; and (5) a professional template library for common book genres. Each step moves the platform closer to a complete, free publishing suite for Bangladesh.

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